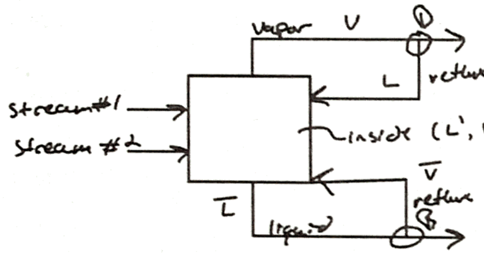
1) PROBLEM 4.D9 10 Rayan Baig

We are separating acetone and ethanol in a distillation column w/ a total condenser and a partial reboiler. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid that is 60.0 mol% acetone & has a feed rate of F1 = 200.0 kmol/h. The second feed is 40.0 mol% acetone, has a feed rate of F2 = 300.0 kmol/h, and is a two-phase mixture that is 20% vapor. Assume CMO is valid. Bottoms product is 10.0 mol% acetone, and distillate product is 80.0 mol% acetone.

A) External reflux ratio $L/D = 2.0$. Find the number of stages required, the optimum feed stage locations, and the boilup ratio $\bar{V}/B$.



TWO FEEDS

$F_1 = 200$ $z_1 = 0.6$
$F_2 = 300$ $z_2 = 0.4$
$x_B = 0.1$ $x_D = 0.8$

(1) $F_1 + F_2 = B + D$
(2) $F_1 z_1 + F_2 z_2 = B x_B + D x_D$

TWO EQNS & TWO unknowns ⇒ solve

⇓

(1) $200 + 300 = B + D$
(2) $200(0.6) + 300(0.4) = B(0.10) + D(0.80)$

$B = 228.571$ kmol/h
$D = 271.429$ kmol/h

TOP OP. line $V = D + L$

$$\text{Slope} = \frac{L}{V} = \frac{L}{D+L} = \frac{L/D}{1+\frac{L}{D}} = \frac{2}{1+2} = \frac{2}{3}$$

Since we know $\frac{L}{D} = 2$

$\frac{L}{D} = 2$

D known from calculation above = 271.429

$L = 2(271.429)\ \frac{\text{kmol}}{h}$

$L = 542.857\ \frac{\text{kmol}}{h}$

$V = D + L$

$V = 271.429 + 542.857$

$V' = V = 814.286\ \frac{\text{kmol}}{h}$

Sat liquid feed

↳ $L' = F_1 + L = 200 + 542.857$

$L' = 742.857\ \frac{\text{kmol}}{h}$

Now For Top op line: $y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$

★ $y = \frac{2}{3}x + \frac{1}{3}(0.80)$ GOES THROUGH $x = y = x_D = 0.80$

Now For Bottom OP Line: $y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$ GOES THROUGH $x = y = x_B = 0.10$

TWO phase feed

$\bar{V} = V' - F_2(y) = V' - 300(0.20) = V' - 60 = 814.286 - 60 = 754.286$

$\bar{L} = L' + F_2(x) = L' + 240 = 742.857 + 240 = 982.286$

$$y = \frac{982.286}{754.286}x - \left(\frac{982.286}{754.286} - 1\right)0.10$$

754.286/228.571

FEED #2: $y = -\frac{0.80}{0.20}x + \frac{1.0}{0.2}(0.40)$

$\boxed{5 \text{ Stages}}$ & $\boxed{\frac{\bar{V}}{B} = 3.30}$

~~opt~~ feed stage?

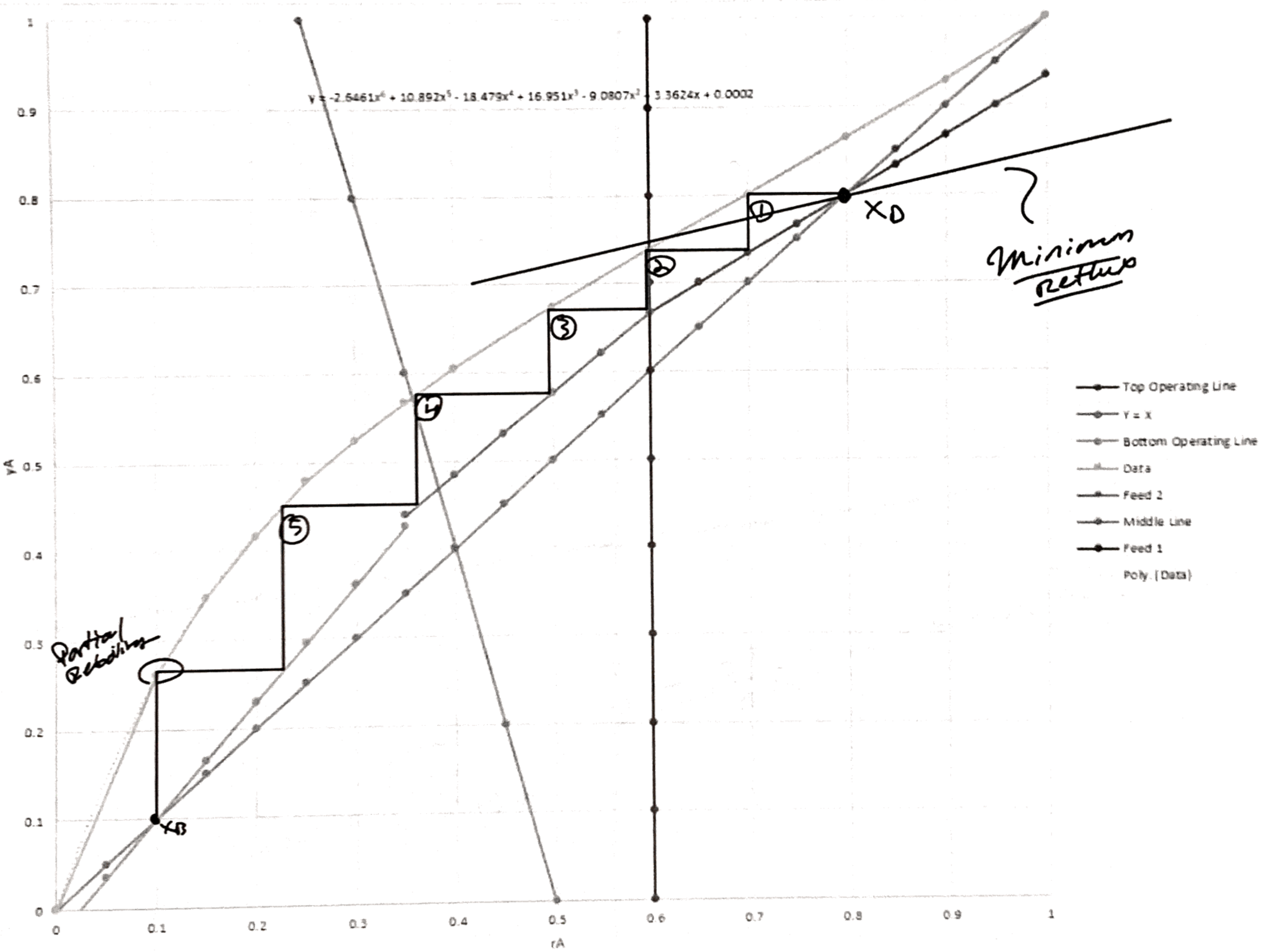
(DATA FROM QUESTION D7 USED)

Find middle line: $y = \frac{L'}{V'}x + \frac{D}{V'}x_D - \frac{F_1 z_1}{V'}$

$$y = \frac{742.857}{814.286}x + \frac{271.429}{814.286}0.8 - \frac{200(0.60)}{814.286}$$

4 D9

Top Operating Line		Bottom Operating Line		Y = X		Data		Feed 1		Feed 2		Middle Line	
X	Y	X	Y	X	Y	xA	yA	X	Y	X	Y	X	Y
0	0.266667	0	-0.03022726	0	0	0	0	0.6	0	0	2	0	0.119298625
0.05	0.3	0.05	0.034886369	0.05	0.05	0.1	0.262	0.6	0.1	0.05	1.8	0.05	0.164912635
0.1	0.333333	0.1	0.1	0.1	0.1	0.15	0.348	0.6	0.2	0.1	1.6	0.1	0.210526645
0.15	0.366667	0.15	0.165113631	0.15	0.15	0.2	0.417	0.6	0.3	0.15	1.4	0.15	0.256140656
0.2	0.4	0.2	0.230227261	0.2	0.2	0.25	0.478	0.6	0.4	0.2	1.2	0.2	0.301754666
0.25	0.433333	0.25	0.295340892	0.25	0.25	0.3	0.524	0.6	0.5	0.25	1	0.25	0.347368676
0.3	0.466667	0.3	0.360454523	0.3	0.3	0.35	0.566	0.6	0.6	0.3	0.8	0.3	0.392982687
0.35	0.5	0.35	0.425568153	0.35	0.35	0.4	0.605	0.6	0.7	0.35	0.6	0.35	0.438596697
0.4	0.533333	0.4	0.490681784	0.4	0.4	0.5	0.674	0.6	0.8	0.4	0.4	0.4	0.484210707
0.45	0.566667	0.45	0.555795414	0.45	0.45	0.6	0.739	0.6	0.9	0.45	0.2	0.45	0.529824718
0.5	0.6	0.5	0.620909045	0.5	0.5	0.7	0.8	0.6	1	0.5	0	0.5	0.575438728
0.55	0.633333	0.55	0.686022676	0.55	0.55	0.8	0.865	0.6	1.1	0.55	-0.2	0.55	0.621052738
0.6	0.666667	0.6	0.751136306	0.6	0.6	0.9	0.929	0.6	1.2	0.6	-0.4	0.6	0.666666749
0.65	0.7	0.65	0.816249937	0.65	0.65	1	1	0.6	1.3	0.65	-0.6	0.65	0.712280759
0.7	0.733333	0.7	0.881363568	0.7	0.7			0.6	1.4	0.7	-0.8	0.7	0.757894769
0.75	0.766667	0.75	0.946477198	0.75	0.75			0.6	1.5	0.75	-1	0.75	0.803508779
0.8	0.8	0.8	1.011590829	0.8	0.8			0.6	1.6	0.8	-1.2	0.8	0.84912279
0.85	0.833333	0.85	1.07670446	0.85	0.85			0.6	1.7	0.85	-1.4	0.85	0.8947368
0.9	0.866667	0.9	1.14181809	0.9	0.9			0.6	1.8	0.9	-1.6	0.9	0.94035081
0.95	0.9	0.95	1.206931721	0.95	0.95			0.6	1.9	0.95	-1.8	0.95	0.985964821
1	0.933333	1	1.272045351	1	1			0.6	2	1	-2	1	1.031578831



Find the minimum external reflux ratio $(L/D)_{min}$ for part a.

Find $\frac{L}{V_{min}}$ first $\therefore$ $\frac{L}{V_{min}} = \frac{x_D - y}{x_D - z_1} = \frac{0.8 - 0.73}{0.8 - 0.6} = 0.35$

$$\left(\frac{L}{D}\right)_{min} = \left(\frac{\frac{L}{V}}{1 - \frac{L}{V}}\right) = 0.538 \qquad \boxed{\left(\frac{L}{D}\right)_{min} = 0.538}$$

c) If feed 2 is changed to a superheated vapor that forces 1.0 mole of liquid to vaporize at the feed stage to cool every 5.0 moles of feed 2 to a saturated vapor, find $(L/D)_{min}$. Hint: The answer is not the same as pt B.

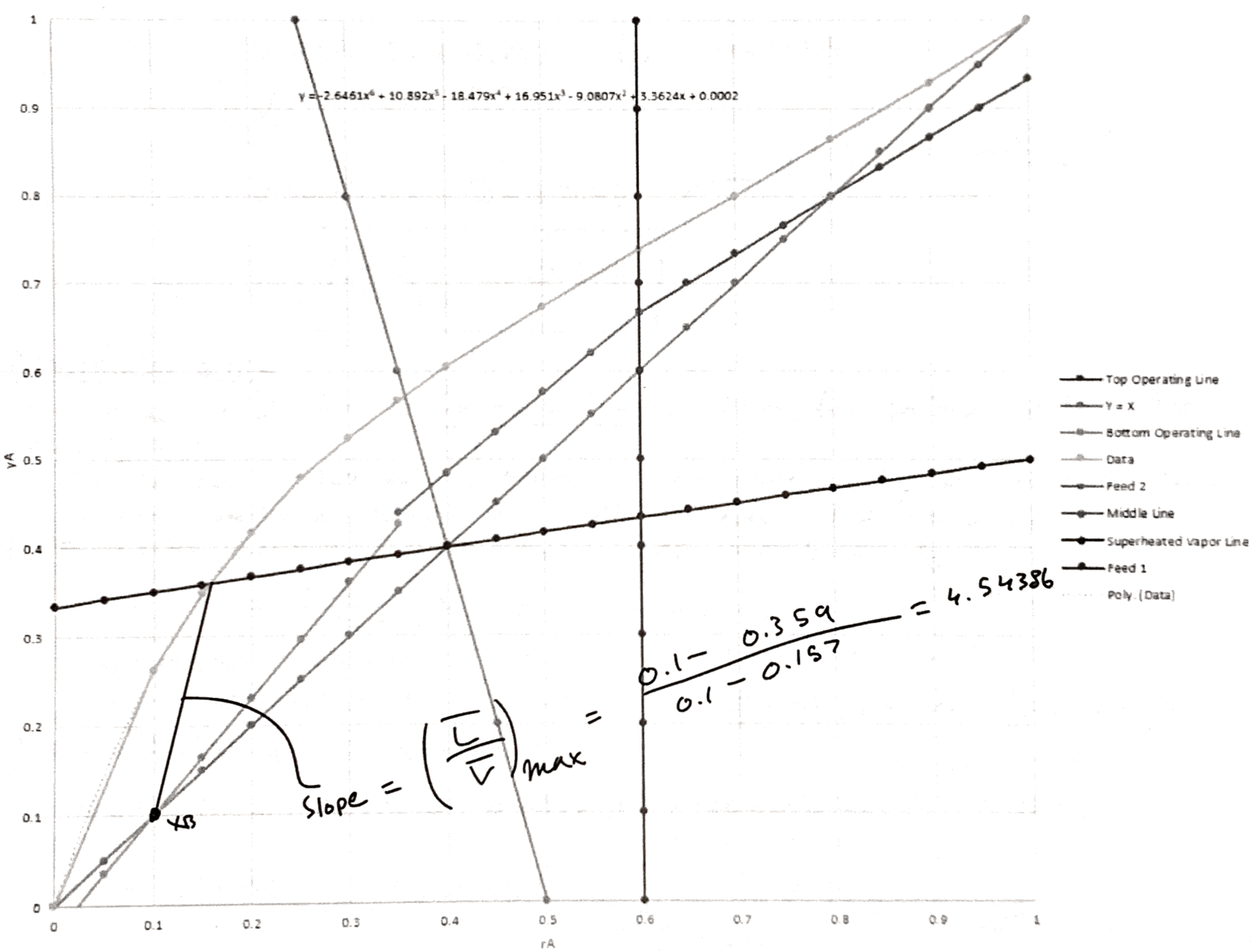
Must Find Superheated vapor line: $\bar{L} = L' - \frac{1}{5}F_2$

$$\text{quality}_{\text{feed 2}} = \frac{\bar{L} - L'}{F_2} = \frac{\cancel{L'} - \frac{1}{5}F_2 - \cancel{L'}}{F_2} = -\frac{1}{5}$$

Slope of feed line: $\frac{q}{q-1} = \frac{0.2}{0.2-1} = -\frac{1}{6}$ — Intersects w/ F_2 line @ $x = y = 0.4$ $\therefore$ Find intercept

$0.4 = \frac{1}{6}(0.4) + b$

$b = \frac{1}{3}$ $\therefore$ $\underbrace{y = \frac{1}{6}x + \frac{1}{3}}_{\text{Find point of intersect w/ data}}$



$$\text{Slope} = \left(\frac{\bar{L}}{\bar{V}}\right)_{max} = \frac{0.1 - 0.359}{0.1 - 0.157} = 4.54386$$

$$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{B}{\bar{V}}\right)x_B$$

$$\left(y + \left(\frac{B}{\bar{V}}\right)x_B\right)\frac{\bar{V}}{x} = \bar{L}$$

$$\left(\frac{\bar{V}}{B}\right)_{min} = \frac{1}{\left(\frac{\bar{L}}{\bar{V}}\right)_{max} - 1} = 0.282$$

$$\frac{\bar{V}}{B} = 0.282$$

$$\bar{V} = (0.282)(228.571)$$

$$\bar{L} = \frac{64.457}{0.157}\left(0.359 + \frac{1}{0.282}\times 0.1\right)$$

$$\bar{L} = 292.976$$

$\bar{V} = 64.457$
$x_B = 0.1$
$x = 0.157$
$y = 0.359$
$\frac{\bar{V}}{B} = 0.282$

$\therefore$

$$\bar{L} = L' - \frac{1}{5}F_2$$

$$L' = \bar{L} + \frac{1}{5}F_2 = 292.976 + \frac{300}{5} = 352.976$$

$$L' = L + F_1 \Rightarrow \quad L = L' - F_1 = 352.976 - 200 = 152.976$$

$$L_{min} = 152.976 \text{ kmol/h}$$

Using

$$\bar{V} = V' - F_2\left(1 + \frac{1}{5}\right)$$

$$V' = V$$

$$V = 64.457 + 300\left(1 + \frac{1}{5}\right) = 424.457$$

$$V_{min} = 424.457$$

$$V = L + D \Rightarrow \quad D = 271.481 \quad \left(\frac{L}{D}\right)_{min} = \frac{152.976}{271.481} = 0.563$$

$$\boxed{\left(\frac{L}{D}\right)_{min} = 0.563}$$

2) PROBLEM 2.D12

Find h_{total} & D for a horizontal flash drum for problem 2.D1C. Use $h_{total}/D = 4$

c) If the feed is 30% mol methanol & we desire a liquid product that is 20% mol methanol, what V/F must be used? $F = 1000.0$ lbmol/h

DATA: (0.30, 0.30) & (0.20, 0.579) OP: $y = -\frac{L}{V}x + \frac{F}{V}z$

$$\text{slope} = -\frac{L}{V} = \frac{0.30 - 0.579}{0.30 - 0.20} = -2.79$$

$$y = -2.79 + \frac{F}{V}z$$

$$0.30 = -2.79(0.30) + b$$

$y_m = 0.579$ $y_w = 1 - 0.579$
$x_m = 0.20$ $x_w = 1 - 0.20$

$$\frac{F}{V}z = b \Rightarrow V = \frac{Fz}{b} = \frac{1000(0.30)}{1.137}$$

$V = 263.85$
$L = 736.148$

Liquid Density using $\rho_L = \frac{\overline{MW_L}}{\bar{V}_L}$

$$\overline{MW}_L = x_M(MW_M) + x_w(MW_w) = (0.20)(32.04) + (0.80)(18.02) = 20.824$$

$$\bar{V}_L = \frac{x_M(MW_M)}{\rho_M} + \frac{x_w(MW_w)}{\rho_w} = \frac{(0.20)(32.04)\text{ g/mol}}{0.792 \frac{g}{mL}} + \frac{(0.80)(18.02)\text{ g/mol}}{1.0 \text{ g/mL}} = 22.5069 \frac{mL}{mol}$$

$$\rho_L = \frac{\overline{MW_L}}{\bar{V}_L} = \frac{20.824 \text{ g/mol}}{22.5069 \frac{mL}{mol}} = 0.925 \text{ g/mL}$$

$\overline{MW}_v = y_m(MW_m) + y_w(MW_w) = 0.574(32.04) + (1-0.574)(18.02) = 26.1376\ g/mol$

$$\rho_v = \frac{P(\overline{MW}_v)}{RT} = \frac{(1.0\,atm)(26.1376\frac{g}{mol})}{(82.0575\,\frac{mL\,atm}{mol\,K})(273.15+81.70)K} = 0.000898\frac{g}{mL}$$

Using ρ_v & ρ_L we can find permissible velocity

$$u = k_{drum}\sqrt{\frac{\rho_L - \rho_v}{\rho_v}}$$

Need

$k_{drum,\,HORIZONTAL} = 1.25\,k_{vertical}$

(EQ 2-67e)

$T = 81.70°C$ $\quad P = 1\,atm$

$A = -1.877478097$
$B = -0.8145804597$
$C = -0.1870744085$
$D = -0.0145228667$
$E = -0.0010148518$

$$k_{vertical} = (constant)\exp[A + B\ln F_{lv} + C(\ln F_{lv})^2 + D(\ln F_{lv})^3 + E(\ln F_{lv})^4]$$

$k_{vertical} = 0.444$

$k_{horizontal} = 1.25 \times k_{vertical}$

$k_{horizontal} = 0.555$

$k_{drum} = constant \rightarrow 1.0\frac{ft}{s}$

$$F_{lv} = \frac{W_L}{W_V}\sqrt{\frac{\rho_v}{\rho_L}}$$

$W_L = L(\overline{MW}_L) = 736.148(20.824)$

* $W_L = 15329.5$

$W_V = V(\overline{MW}_v) = 263.85(26.1376)$

* $W_V = 6896.41$

$$F_{lv} = \frac{15329.5}{6896.41}\sqrt{\frac{0.000898}{0.925}}$$

$$u = k_{drum}\sqrt{\frac{\rho_L - \rho_v}{\rho_v}}$$

$$u = 0.555\sqrt{\frac{0.925 - 0.000898}{0.000898}}$$

$u_{perm} = 17.81\,ft/s$

$$A_c = \frac{V(\overline{MW}_v)}{u_{perm}(3600\rho_v)} = \frac{250(26.1376)}{17.81\frac{ft}{s}(3600 \times 0.000898)} \times \underbrace{\frac{454\,g/lbm}{28316.85\,\frac{mL}{ft^3}}}_{Converting} = 1.82\,ft^2$$

[HORIZONTAL]

$0.2A_{total} = A_c \Rightarrow A_c = 9.0956\,ft^2$

$$\text{Now using } D = \sqrt{\frac{4A_T}{\pi}} = \sqrt{\frac{4(9.14752\,ft^2)}{\pi}} = 3.403\,ft$$

since $\frac{h_{total}}{D} = 4$

$h_{total} = 13.612\,ft$

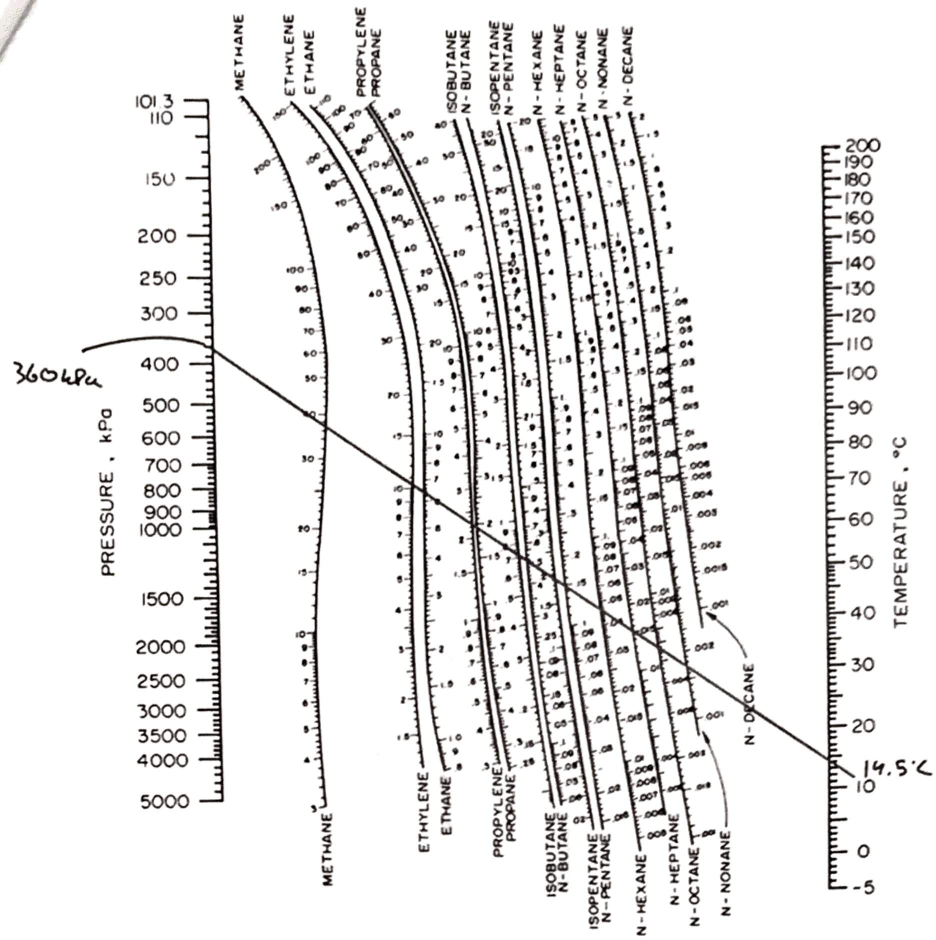
$\boxed{D = 3.40\,ft}$ & $\boxed{h = 13.6\,ft}$

3) Problem 2.D13

We flash distil a mixture that is 36% ethane (C_2) & 64% n-butane C_4. The flash drum operates as an equilibrium stage. We measure the outlet concentrations of ethane as $x_{C_2} = 0.088$ & $y_{C_2} = 0.546$. Find x_{C_4}, y_{C_4}, T_{drum}, P_{drum} & $\frac{V}{F}$.
(NO TRIAL & ERROR)

#1 → ethane $\quad z_1 = 0.36 \quad x_{C_2} = 0.088 \quad y_{C_2} = 0.546 \quad K_{C_2} = \frac{0.546}{0.088} = 6.20$

#2 → n-butane $\quad z_2 = 0.64 \quad x_{C_4} = 0.912 \quad y_{C_4} = 0.454 \quad K_{C_4} = \frac{0.454}{0.912} = 0.498$



$T_{drum} = 14.5°C$

$P_{drum} = 360$ kPa

Find $\frac{V}{F}$ w/ Mass Balance

$F = L + V$, $Fz = Lx + Vy$

$L = F - V$

$$\frac{Fz = (F-V)x + Vy}{F}$$

$$z = \left(1 - \frac{V}{F}\right)x + \frac{V}{F}y$$

$$z - x = \frac{V}{F}(-x + y)$$

$$\frac{V}{F} = \frac{z - x}{y - x}$$

$$\frac{V}{F} = \frac{0.36 - 0.088}{0.546 - 0.088}$$

$\boxed{\frac{V}{F} = 0.594}$

$T_{drum} = 14.5°C$

$P_{drum} = 360$ kPa

$x_{C_4} = 0.912$

$y_{C_4} = 0.454$

4) PROBLEM 2.D14

A flash drum is separating a mixture that is 12.0 mol% methane C_1, 48.0% and n-Butane C_4, & 40.0% n-pentane C_5. Feed rate is 122.0 $\frac{kmol}{h}$. The feed is partially liquid & partially vapor at a pressure of 5.0 bar & temperature of 50.4°C. The flash drum is at 3.0 bar and $T = 36°C$. Find $\frac{V}{F}$, the k values, and vapor & liquid mole fractions.

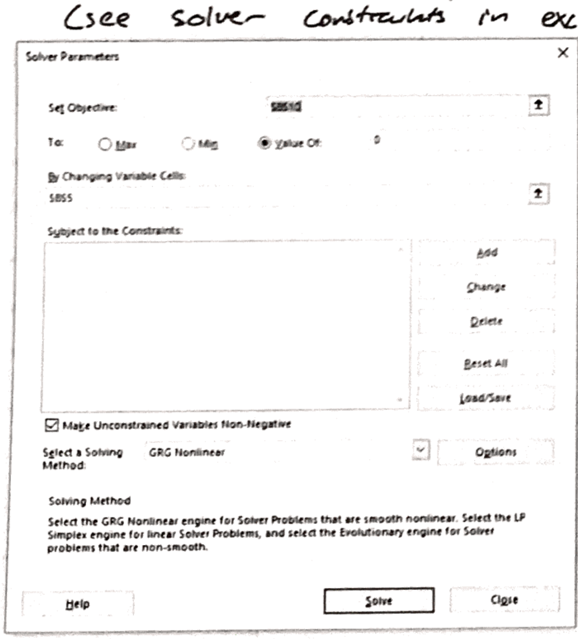
#1 C_1 $z_1 = 0.12$ $k_1 = 50$

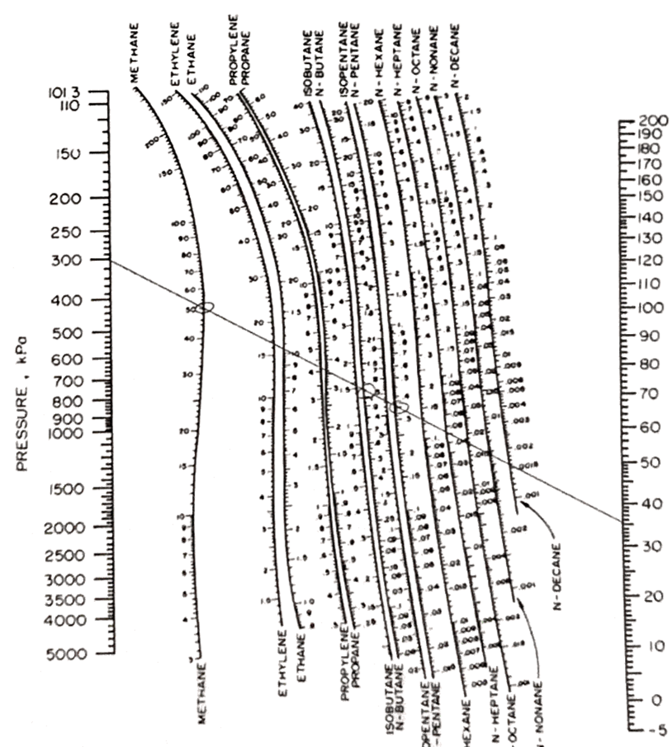
#2 C_4 $z_2 = 0.48$ $k_2 = 1.1$

#3 C_5 $z_3 = 0.40$ $k_3 = 0.36$

RR equation, solve for $\frac{V}{F}$

(see solver constraints in excel)





	A	B	C	D
1	Problem 2.D13			
2	Components in Mixture:	Methane	N-Butane	N-Pentane
3	K Values:	50	1.1	0.36
4	Molar Composition:	0.12	0.48	0.4
5	V/F:	0.51137		
6	Liquid Composition:	0.00461	0.456648	0.594599
7	Vapor Composition:	0.23026	0.502313	0.214056
8				
9	Equations	LHS	RHS	
10	Rachford-Rice Equation:	-0.1092	0	

w/ initial Guess

	A	B	C	D
1	Problem 2.D13			
2	Components in Mixture:	Methane	N-Butane	N-Pentane
3	K Values:	50	1.1	0.36
4	Molar Composition:	0.12	0.48	0.4
5	V/F:	0.38747		
6	Liquid Composition:	0.006	0.462095	0.531901
7	Vapor Composition:	0.30021	0.508305	0.191484
8				
9	Equations	LHS	RHS	
10	Rachford-Rice Equation:	4.8E-13	0	

Solver finds a solution

$$\boxed{\begin{aligned} \frac{V}{F} &= 0.511 \\ X_1 &= 0.006 \\ X_2 &= 0.462 \\ X_3 &= 0.532 \\ Y_1 &= 0.300 \\ Y_2 &= 0.508 \\ Y_3 &= 0.191 \end{aligned}}$$